

Homework 8

- **Submit your solutions electronically on the course Gradescope site as PDF files.** If you plan to typeset your solutions, please use the $\text{\LaTeX}$ solution template on the course web site. If you must submit scanned handwritten solutions, please use a black pen on blank white paper and a high-quality scanner app (or an actual scanner, not just a phone camera). We will mark difficult to read solutions as incorrect and move on.
- **Every homework problem must be done *individually*.** Each problem needs to be submitted to Gradescope before midnight of the due date which can be found on the course website: <https://ecealgo.com/su26/homeworks.html>.
- For nearly every problem, **we have covered all the requisite knowledge required to complete a homework assignment prior to the “assigned” date.** This means that there is no reason not to begin a homework assignment as soon as it is assigned. Starting a problem the night before it is due is a recipe for failure.

Policies to keep in mind

- **You may use any source at your disposal**—paper, electronic, or human—but you *must* cite *every* source that you use, and you *must* write everything yourself in your own words. See the academic integrity policies on the course web site for more details.
- **Being able to clearly and concisely explain your solution is a part of the grade you will receive.** Before submitting a solution ask yourself, if you were reading the solution without having seen it before, would you be able to understand it within two minutes? If not, you need to edit. Images and flow-charts are very useful for concisely explain difficult concepts.

See the course web site (<https://ecealgo.com/su26>) for more information.

If you have any questions about these policies,
please don't hesitate to ask in class, in office hours, or on Piazza.

1. A Hamiltonian cycle in a graph is a cycle that visits every vertex exactly once. A Hamiltonian path in a graph is a path that visits every vertex exactly once, but it need not be a cycle (the last vertex in the path may not be adjacent to the first vertex in the path.) Consider the Directed Hamiltonian Cycle problem, which checks whether a Hamiltonian cycle exists in a directed graph, the Undirected Hamiltonian Cycle problem, which checks whether a Hamiltonian cycle exists in an undirected graph, and the Undirected Hamiltonian Path problem, which checks whether a Hamiltonian path exists in an undirected graph.
 - Give a polynomial time reduction from the *directed* Hamiltonian cycle problem to the *undirected* Hamiltonian cycle problem.
 - Give a polynomial time reduction from the *undirected* Hamiltonian Cycle to *directed* Hamiltonian cycle.
 - Give a polynomial-time reduction from undirected Hamiltonian *Path* to undirected Hamiltonian *Cycle*.
2. For each of the following problems, pick true or false and explain why.
 - (a) **True/False:** If L is an NP-complete language and $L \in P$, then $P = NP$.
 - (b) **True/False:** There exists a polynomial-time reduction from every problem in NP to every problem in P.
 - (c) **True/False:** If a problem is both NP-hard and co-NP-hard, then it must be in NP.
 - (d) **True/False:** If there is a polynomial-time reduction from problem A to problem B and B is in NP, then A must also be in NP.
 - (e) **True/False:** If a problem is solvable in polynomial space, then it is also solvable in polynomial time.
3. You are given a set U of n elements, a collection of m subsets $S_1, S_2, \dots, S_m$, where each $S_i \subseteq U$, and a positive integer k . The Set Packing problem asks whether there exists a collection of at least k subsets such that no two selected subsets share any elements.

Prove that Set Packing is NP-hard. Give a polynomial-time reduction from the Independent Set problem to Set Packing and explain why your reduction is correct.